‘Don’t worry, if something goes wrong, you will not die alone. I stay here with you.’

I am not sure if it was any reassurance, but Thankaiyan later told me that he saw in my eyes a determination that he dared not question.

I summoned the security chief and instructed him to evacuate the place, basically send out all people from the vicinity of the test facility.

‘If anyone comes into the building, you will go out forever,’ I told the security chief. ‘Don’t let in any visitor, even if he says he is the chairman of ISRO.’

Thankaiyan and I got to work, he deftly drilling, and I quenching the spot with wet cotton to minimise, if not prevent, sparks. The job was done in an hour, and the engine was test-ready, minus the transducer.

Menon came as scheduled. We fired the engine. The test was a success.

SEP wouldn’t have allowed any such adventure. Here, protocol was supreme and the different teams worked in tandem. Egos never interfered with coordination, at least not in the open. But the best of protocols have holes; and a natural corollary to this are mistakes that often result from human carelessness. What else explains the failure of a test that was finally attributed to a worker leaving cotton waste in a rocket part? The protocol was to plug all the orifices—and there are some 500 of them—of the rocket engine every day after work. This is to prevent any impurity from getting into the feeder lines or the combustion chamber. A worker, while covering these openings with cotton, used lesser quantity at one place, and the cotton slipped in. Nobody checked before the engine was taken for test firing.

The ISRO team took copious notes of such mistakes and of high-tech lessons we learned. While the daily meetings were meant for discussing what each team worked on, the team leaders were asked to present problems they did not have answers to, at the weekly meetings. Not that all unanswered questions meant inefficiency of the engine—the engine may work fine even if you do not know how a certain material would react if the operating pressure goes up beyond a point—but finding the answers was crucial to the know-why. Questions led to analysis and eventually answers.